

Build the complete delivery course

Name: _____ Team: _____ Date: _____

A robot that carries a parcel into a marked delivery space.

Gather these materials

- ☐ Checked base and the selected holder from lesson 3
- ☐ The same light paper parcel
- ☐ White lane and red strip from lesson 6
- ☐ Removable tape, ruler and saved code

Build: steps 1-4

- ☐ 1. Turn the robot off. Attach the tested holder to the approved mounting points. Put the paper parcel low in the center. Check that no part touches a wheel or covers the sensor.
- ☐ 2. Keep the white approach and red strip from lesson 6. Tape a delivery rectangle around the stopping area, large enough for the whole robot.
- ☐ 3. Keep the lane clear. For this test, use a short straight approach to the red strip. Add a longer route only after this works.
- ☐ 4. Use the checked color-stop program from lesson 6. Keep its short moves, Stop steps and maximum of 20 checks.

Before power: parts secure, wheels clear, Stop controls easy to reach.

Before a run: clear the lane and ask your teacher to check the setup.

After a run: stop and turn off the robot before changing any parts.

Drawings show the idea. Measure your own pieces; drawings are not full size.

Finish, code and test

Name: _____ Team: _____ Date: _____

Build: steps 5-8

[] 5. Write three rules before driving: parcel stays on; robot stops inside the delivery space; robot hits nothing. A run passes only when all three happen.

[] 6. Run the delivery three times from the same start. Write Yes or No for each rule after each run. Keep the failed runs in the notes.

[] 7. Choose one change, such as a lower holder wall or wider red strip. Keep the other parts and code the same. Try another set of three runs.

[] 8. Save the best-tested setup. Keep the base ready for the final build choice.

Make the program

Use the complete lesson 6 color-stop program.

White permits one 0.1-second slow move and then Stop. Red or any other reading ends the test.

Stop and end after at most 20 checks. Never restart within the same test.

Teacher check: secure attachment, free wheels, clear sensor, reachable Stop controls, safe lane. Turn the power off before changing parts.

Record your test runs

Run	Parcel stayed on?	Inside space?	Hit nothing?
1			
2			
3			

A successful build

At least two of three runs meet all three rules.

The robot still passes the red-and-unknown stop checks.

The team can name one change and the result it produced.

What improved when you changed one part of the delivery?